Metabolic Pathway Maps

Practical, Nicolas Huber

Supervisors: Prof. Bas Teusink, Dr. Brett Olivier

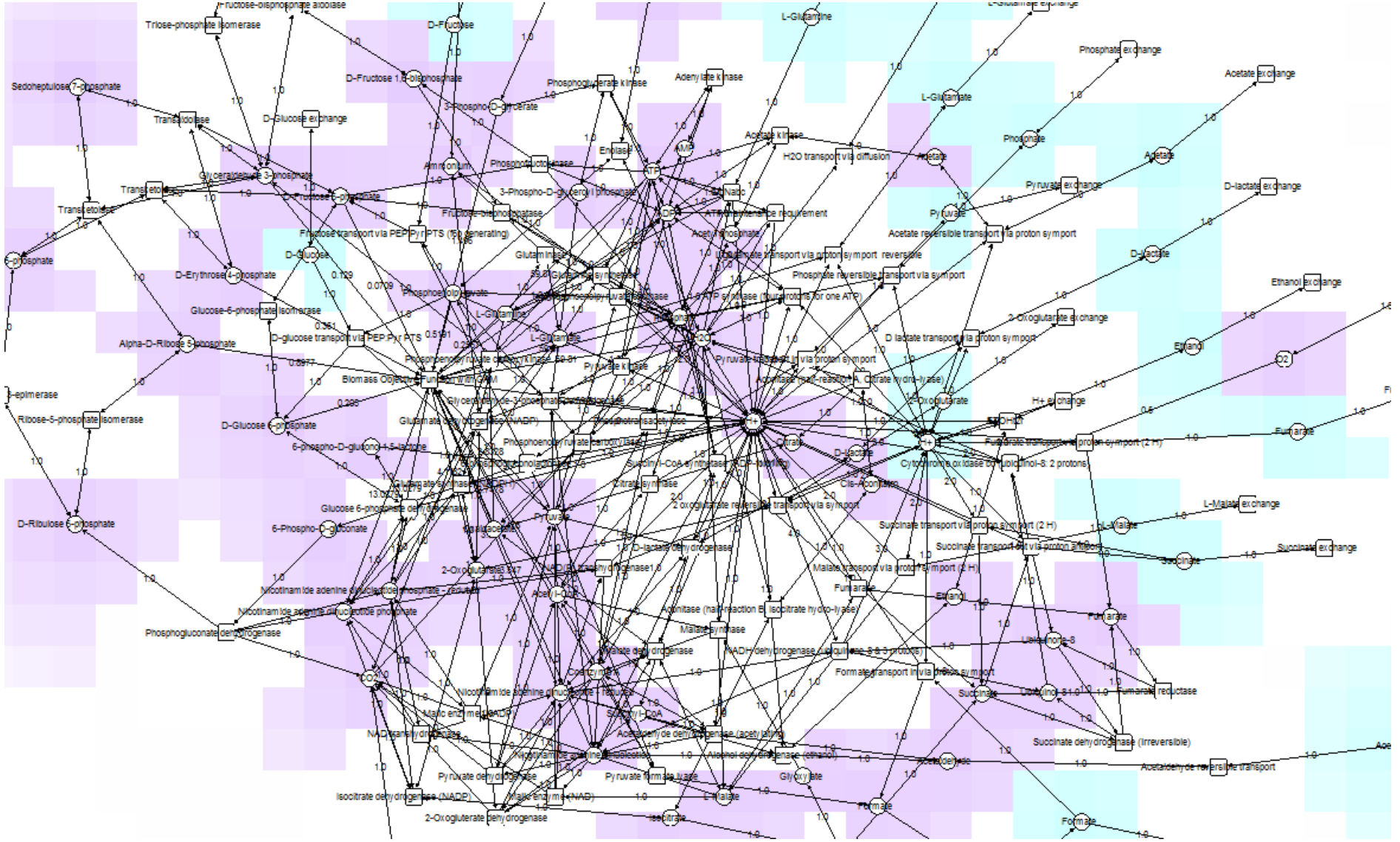
Why develop another tool?

	Structured Layout	Scalability	Data Overlay	SBML	Free	Open Source	Standalone	Maintenance
Acorn	yes	no	yes	yes	yes	yes	no	no
Arcadia	yes	no	no	yes	yes	yes	yes	no
Cytoscape	no	yes	yes	no	yes	yes	yes	yes
FAME	yes	no	yes	yes	yes	no	no	yes
GEMSiRV	yes	no	yes	no	yes	no	yes	no
MetDraw	no	Yes	yes	yes	yes	yes	yes	yes
Mimoza	yes	no	no	yes	yes	yes	yes	yes
openCOBRA	yes	no	yes	no	no	yes	yes	yes
OptFlux	no	yes	no	no	yes	yes	yes	yes
PathVisio	yes	no	yes	no	yes	no	yes	yes
PathwayTools	yes	yes	yes	no	yes	no	yes	yes
Raven	yes	yes	yes	yes	no	no	yes	no
Vanted	no	yes	yes	yes	yes	yes	yes	yes

Vanted

E. coli, core model

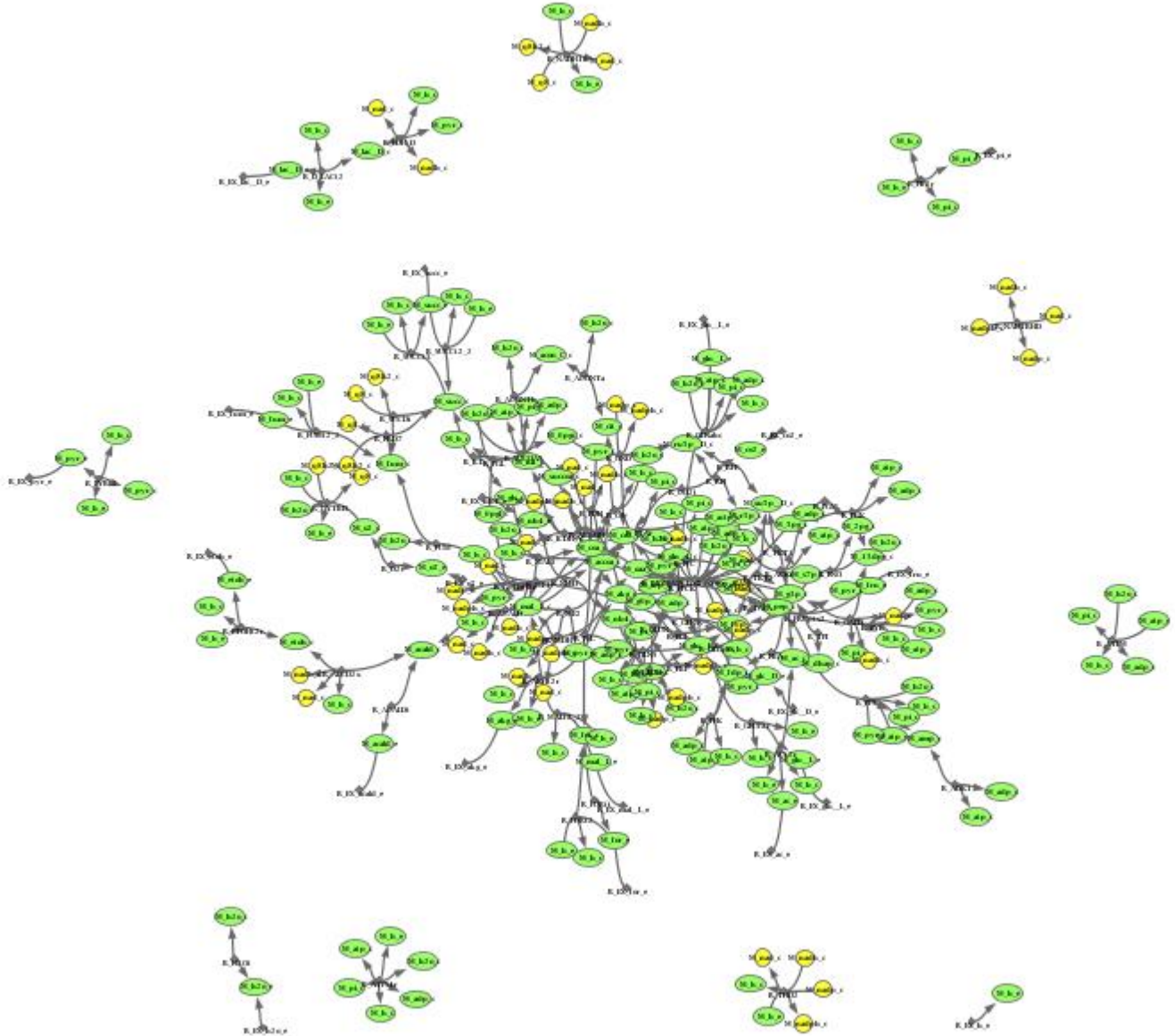
72 species



OptFlux

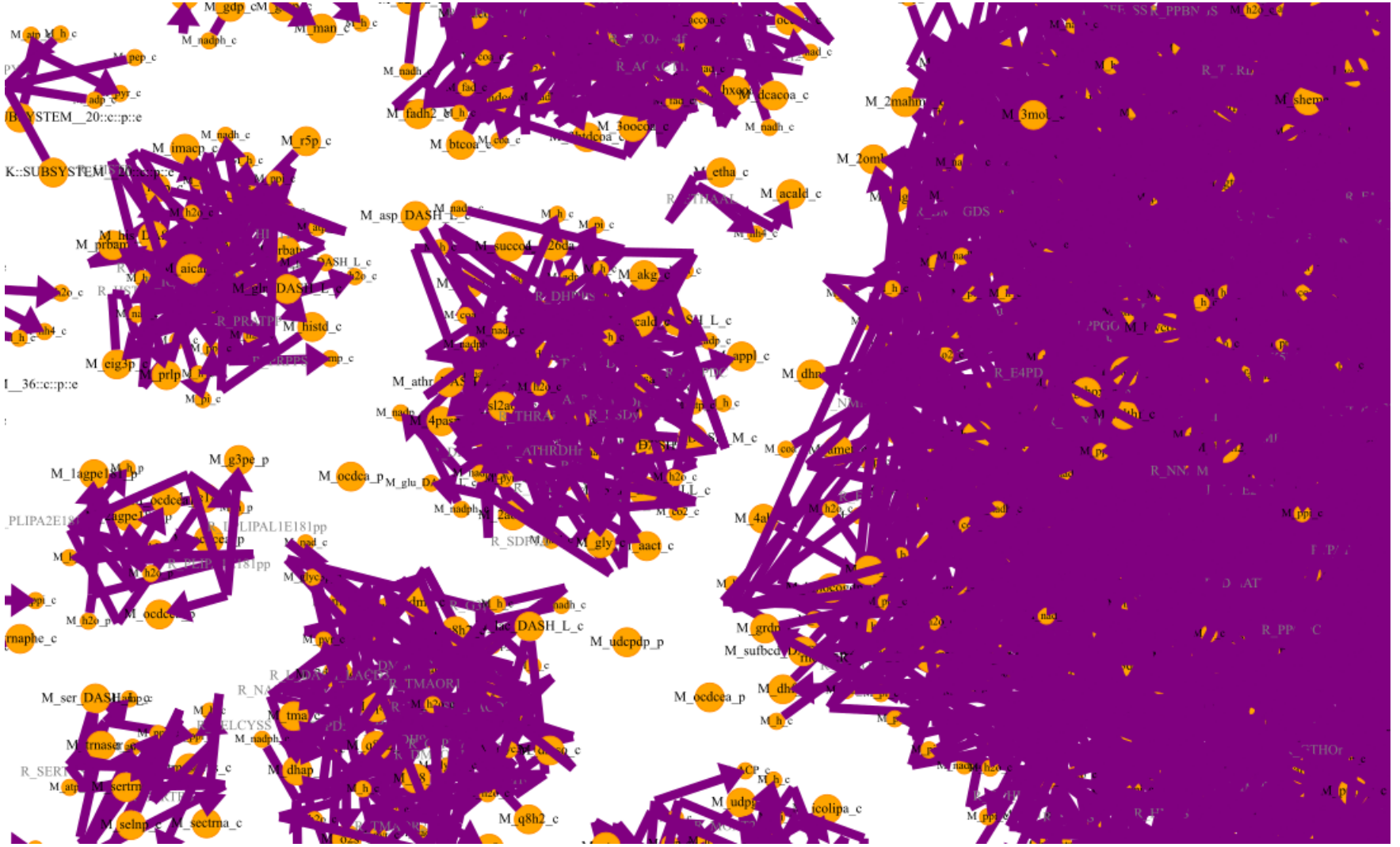
E. coli, core model

72 species



MetDraw

E. coli, genome-scale model
1367 species



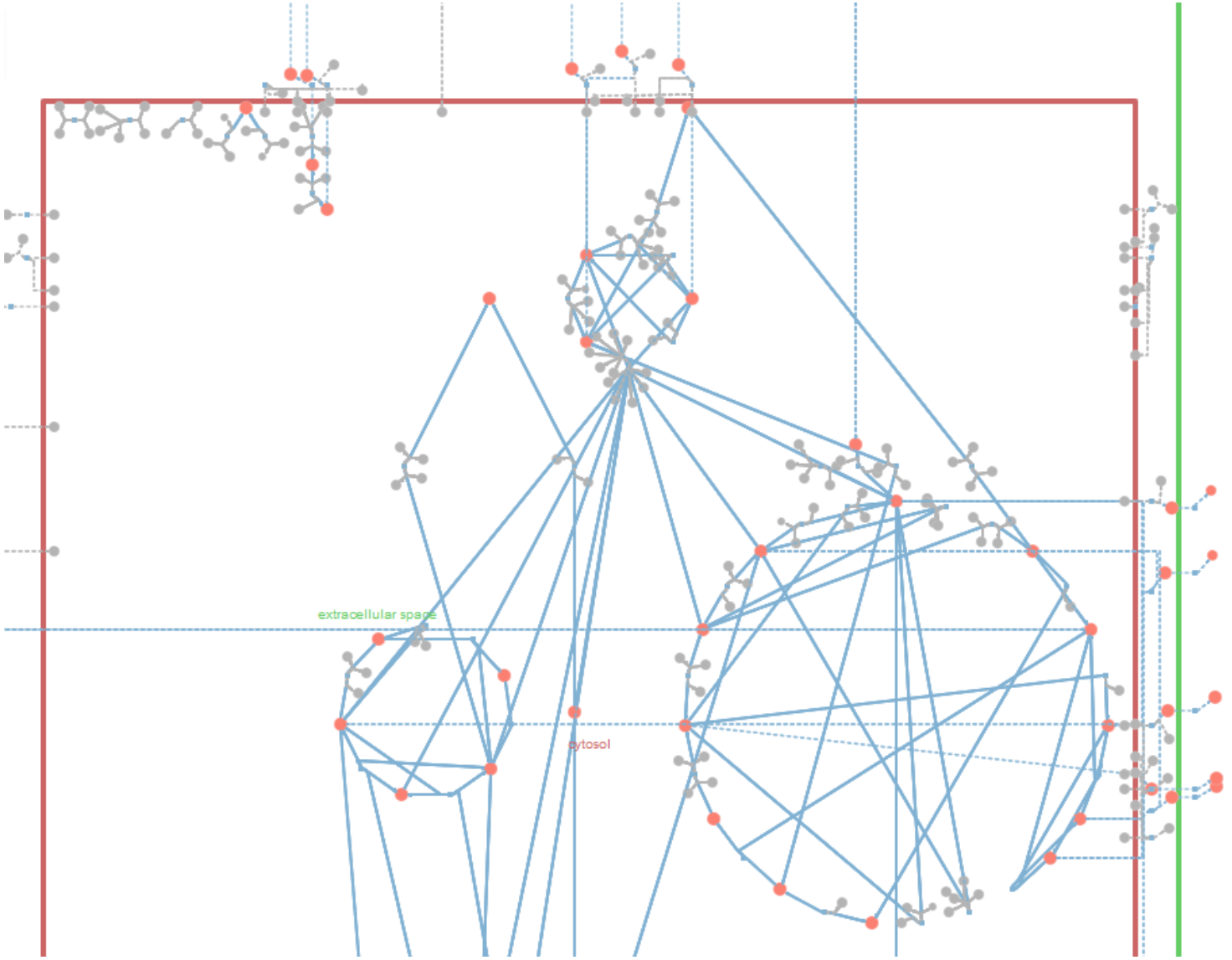
automatic layout algorithms generate messy "hairballs"!

<->

manual layouting takes away valuable time!

Mimoza

E. coli, core model
72 species



automatically generated layout
that structures the network graph!

→

how does the algorithm work?
also possible for larger, genome-scale models?

Mimoza uses the Tulip Python API:

"Tulip is an information visualization framework written in C++ dedicated to the analysis and visualization of graphs."

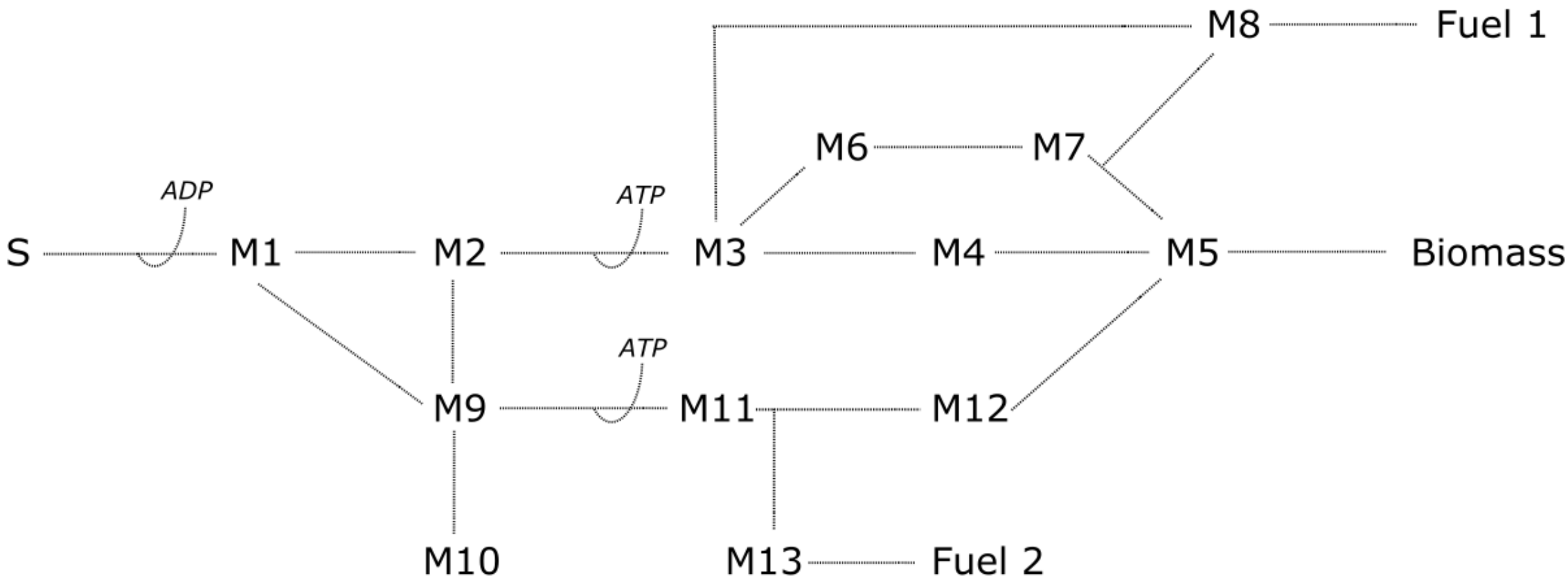


```
30  def layout_hierarchically(qo, margin=1):
31      root = qo.getRoot()
32      ds = tlp.getDefaultPluginParameters(HIERARCHICAL_GRAPH, qo)
33      if qo.numberOfNodes() > 1:
34          # looks like there is a bug in Tulip and it uses the 'layer spacing' value
35          # instead of the 'node spacing' one and visa versa
36          d = SPECIES_SIZE + margin #get_distance(qo)
37          ds["layer spacing"] = d
38          ds["node spacing"] = d
39          ds["layer distance"] = d
40          ds["node distance"] = d
41      qo.applyLayoutAlgorithm(HIERARCHICAL_GRAPH, root[VIEW_LAYOUT], ds)
42
43
44  def layout_circle(qo, margin=1):
45      root = qo.getRoot()
46      ds = tlp.getDefaultPluginParameters(CIRCULAR, qo)
47      if qo.numberOfNodes() > 1:
48          dist = get_distance(qo) + margin
49          ds["minDistCircle"] = dist
50          ds["minDistLevel"] = dist
51          ds["minDistCC"] = 1
52          ds["minDistSibling"] = dist
53      qo.applyLayoutAlgorithm(CIRCULAR, root[VIEW_LAYOUT], ds)
54
55
56  def layout_force(qo, margin=1):
57      root = qo.getRoot()
58      ds = tlp.getDefaultPluginParameters(FM3, qo)
59      ds["Unit edge length"] = 2 * SPECIES_SIZE + margin
60      qo.applyLayoutAlgorithm(FM3, root[VIEW_LAYOUT], ds)
61
62      remove_overlaps(qo, margin)
```

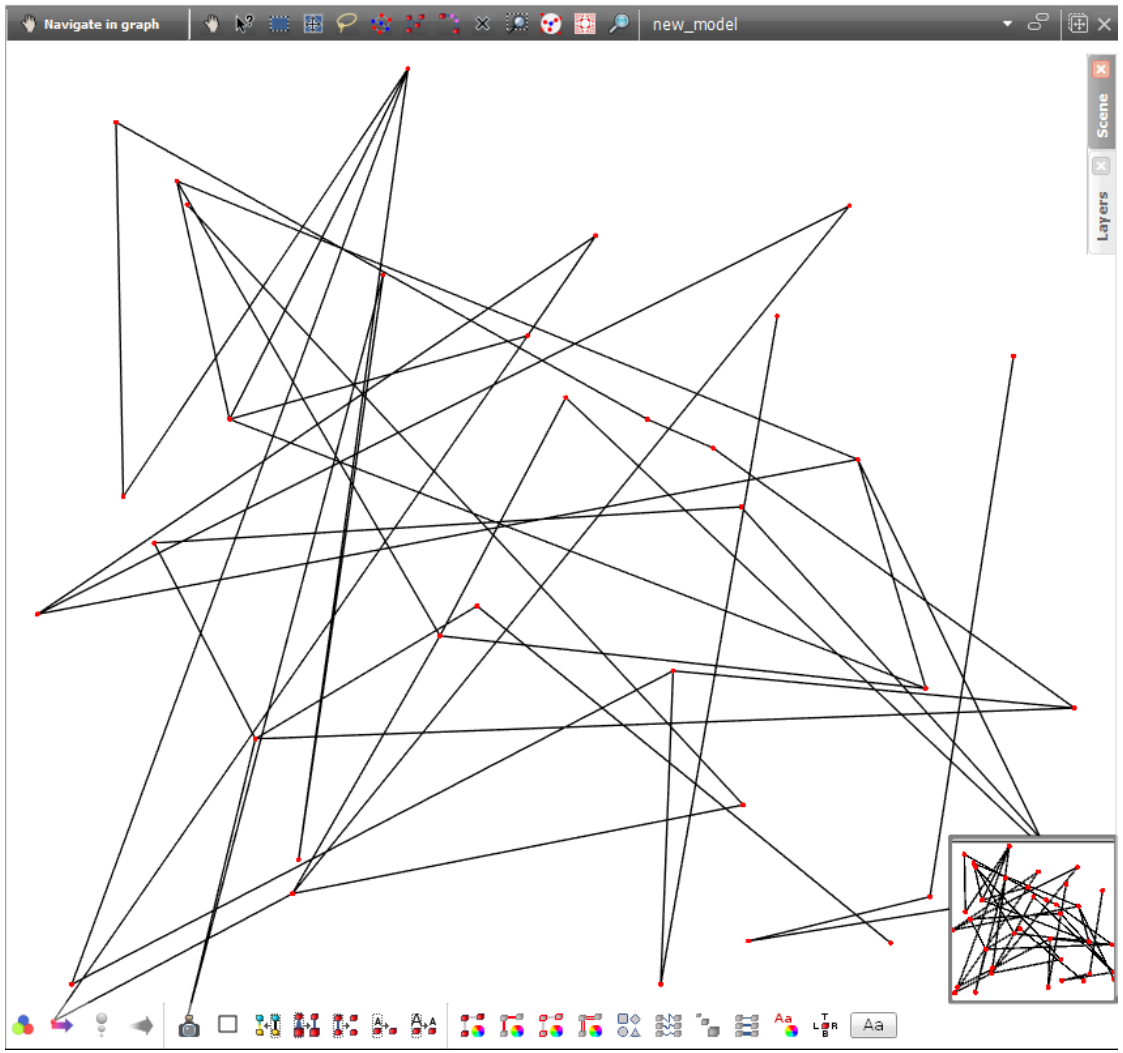
```
99      # iterate over strongly connected components
100     for cc in tarjan_iter({n: list(follower_method(n, gr)) for n in ns}):
101         if len(cc) > 3:
102             if dfs(cc[0], gr, set(), None, cycle_number_threshold, follower_method) \
103                     <= cycle_number_threshold:
104                 mn = gr.createMetaNode(cc, False)
105                 cc_graph = root[VIEW_META_GRAPH][mn]
106                 layout_circle(cc_graph, margin)
107                 meta_ns.append(mn)
108             else:
109                 process_cc(gr, cc, meta_ns, follower_method_iterator)
110
```

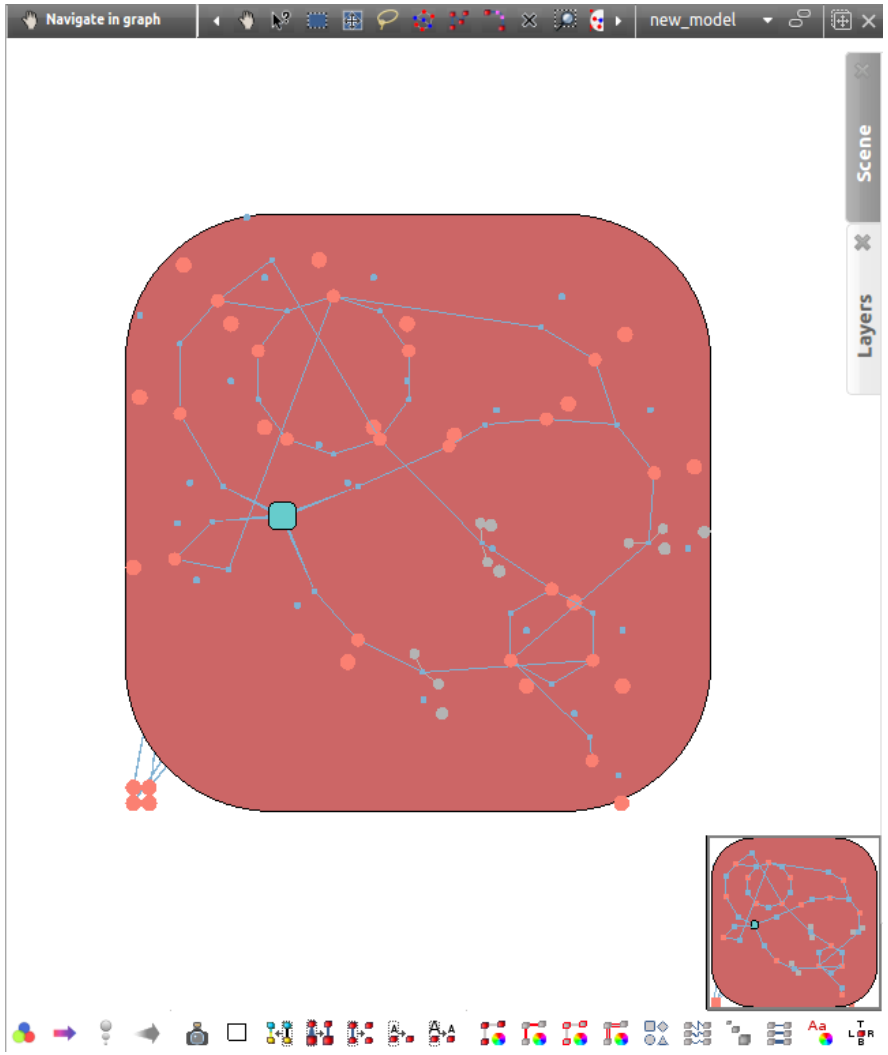
Tulip Layout Algorithms … are called under certain conditions (connectivity)

example: toymodel biofuel
19 species

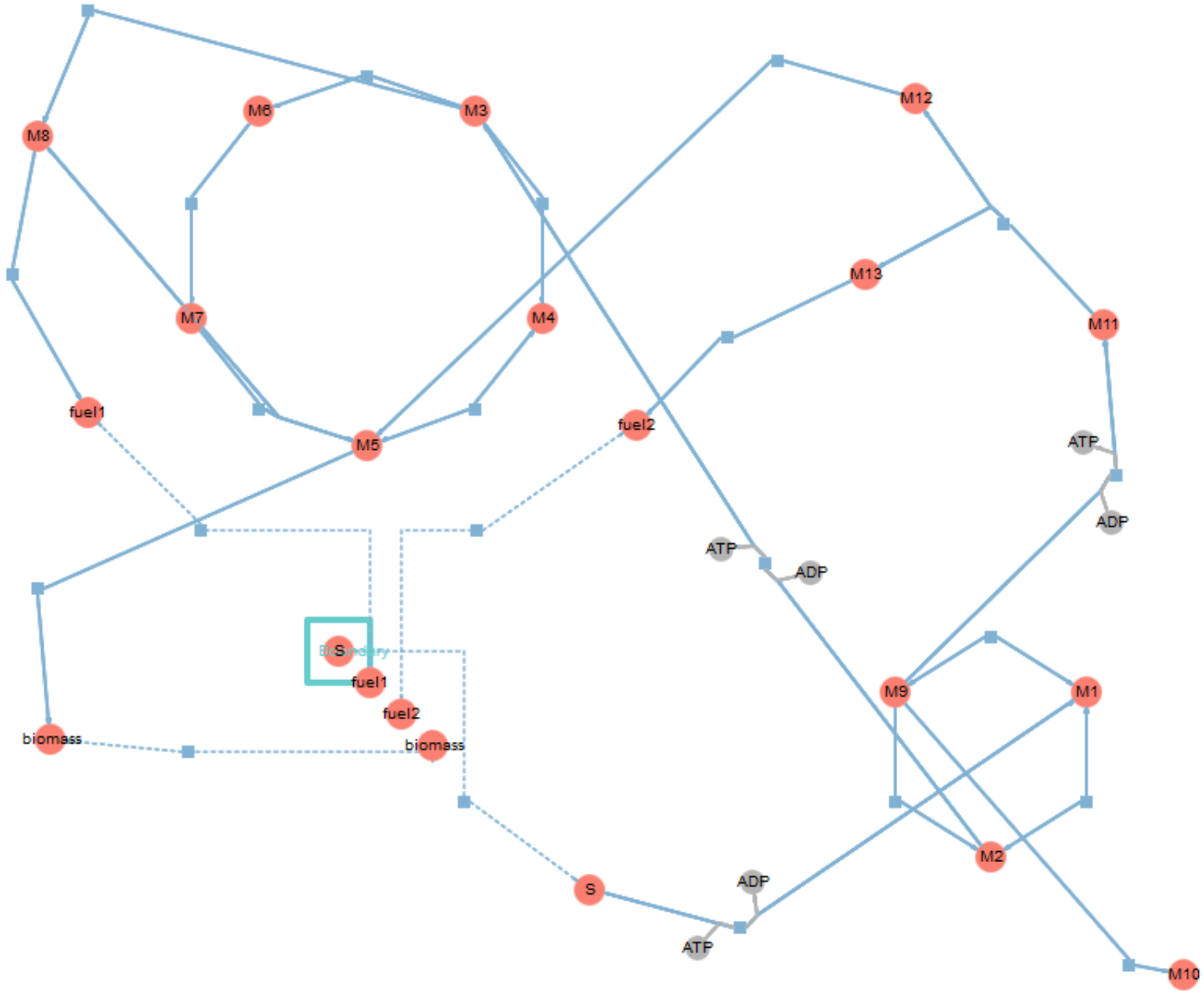


example: toymodel biofuel
19 species



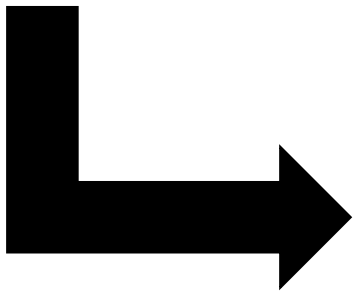


example: toymodel biofuel
19 species



(0. Learn more about Python, Git, SBML Level 3) ✓

1. Find and test existing model visualization tools ✓
2. Understand the Mimoza source code ✓
3. Extract the layouting part ✓



4. Write code that takes **layouting basics from Mimoza**

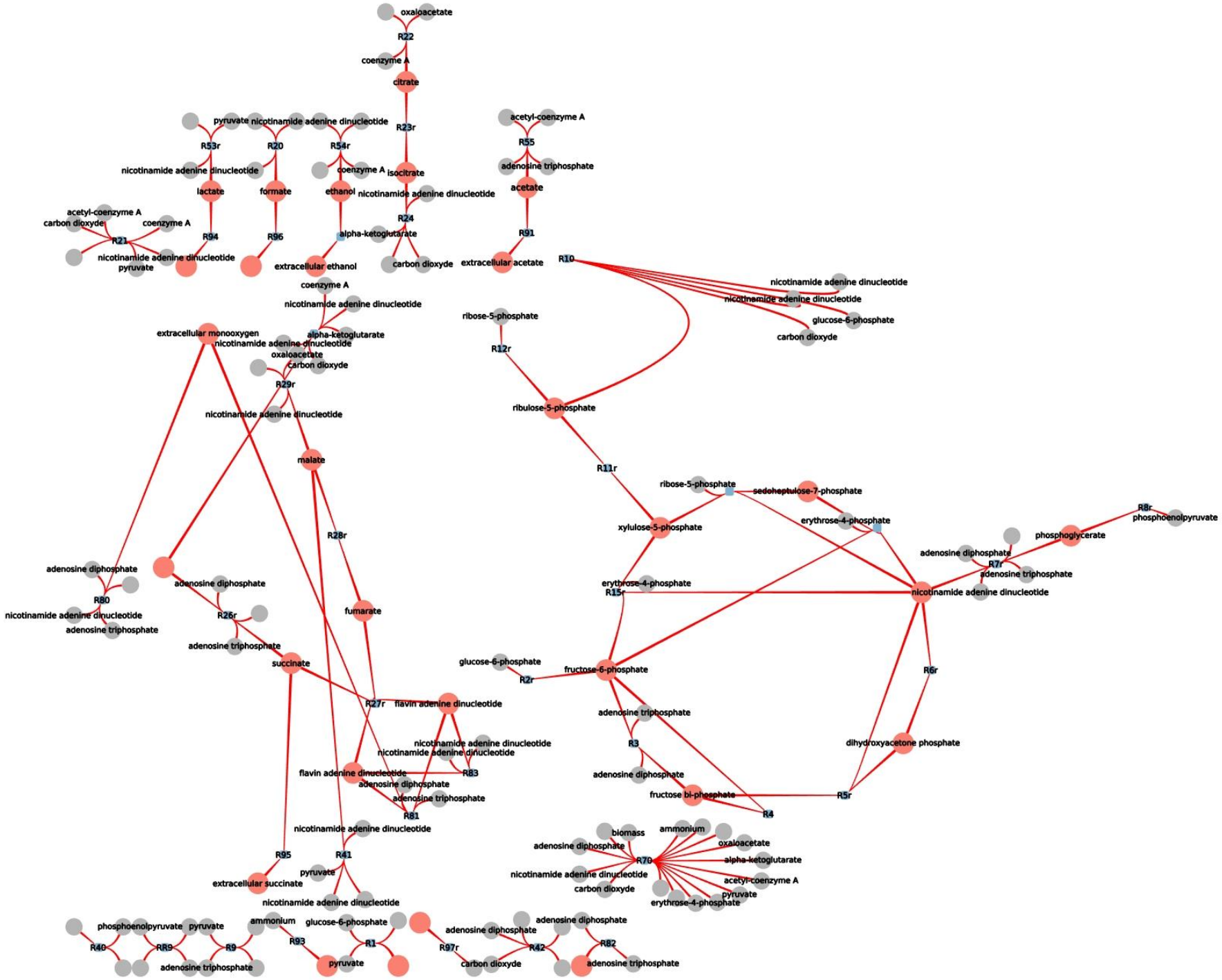
   and applies **additional Tulip functions**

   to **generate structured layout** even for large, genome-scale models

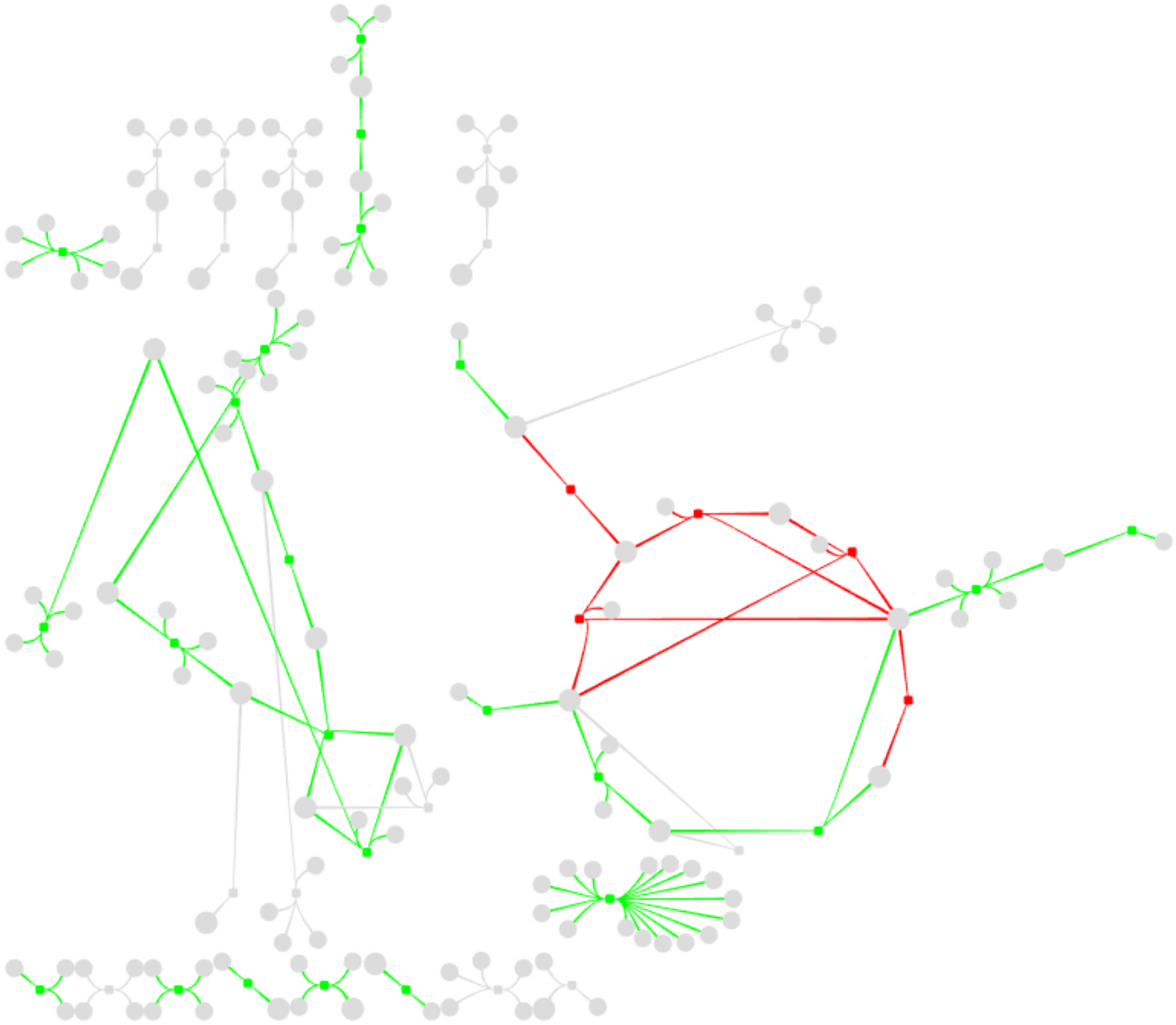
   (+ additional functionality such as overlaying FBA results, etc.!)

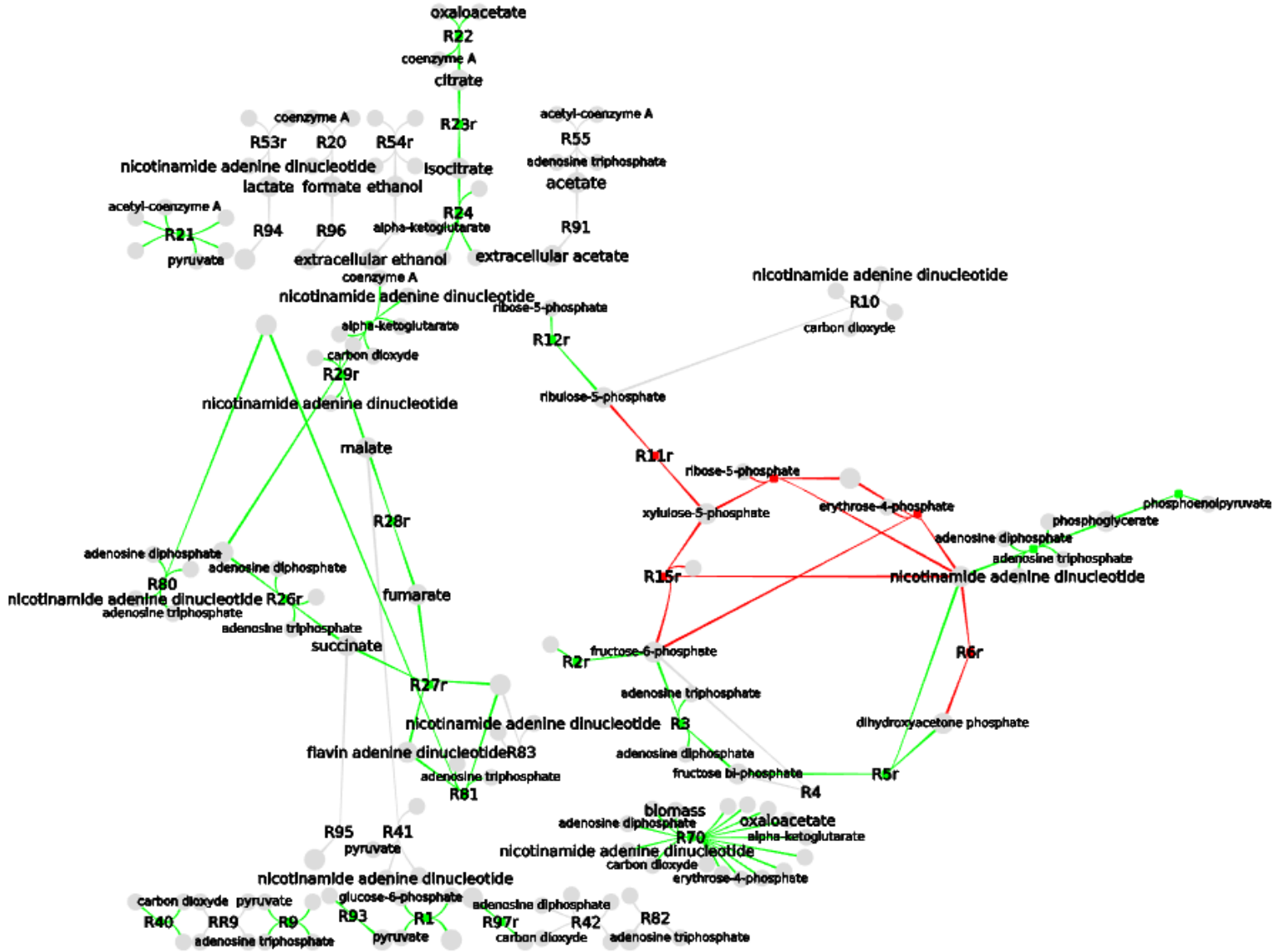
(✓)

E. coli, core model
Carlson, 2003

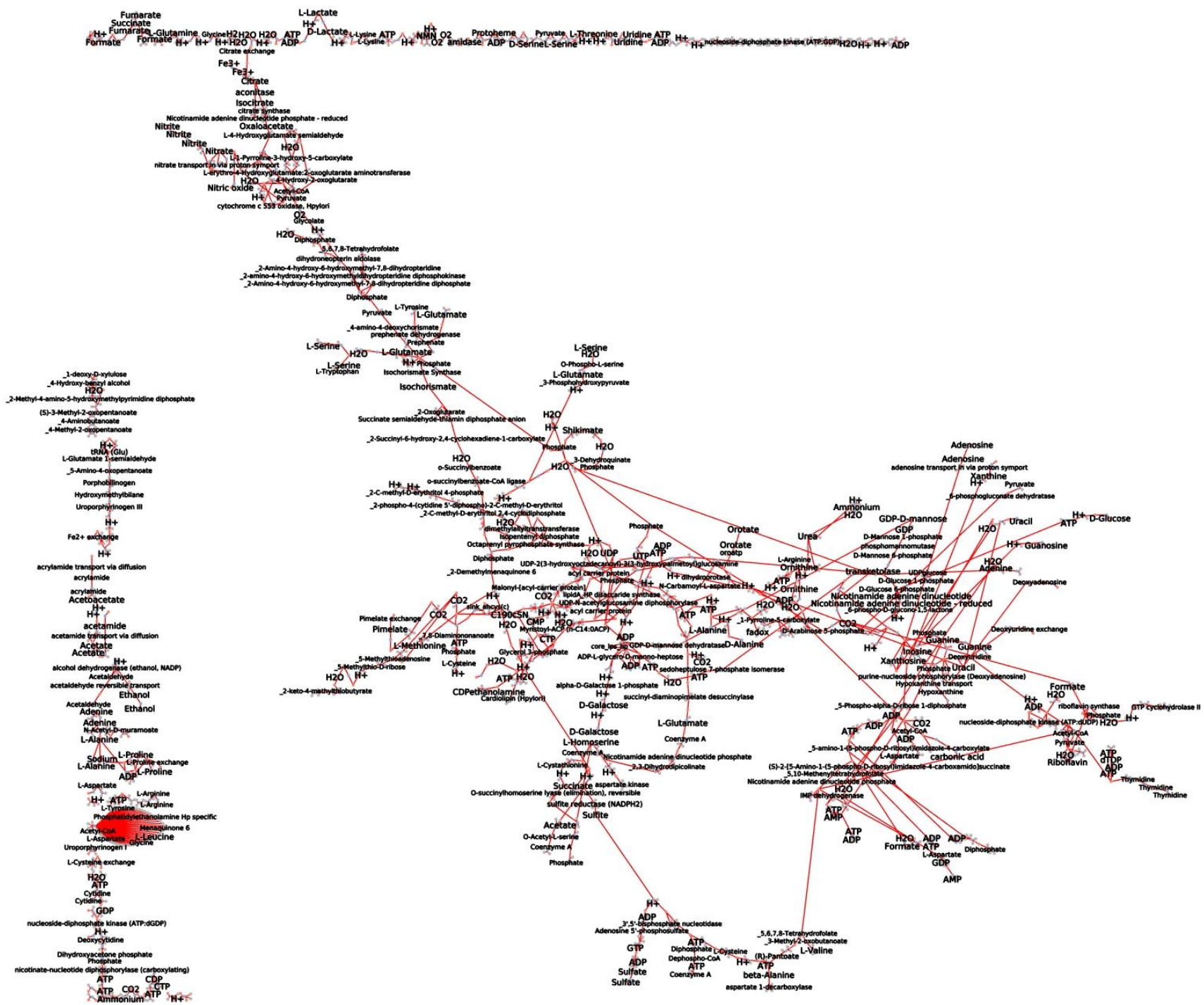


FBA results (with cbmpy):
green = positive flux
red = negative flux
grey = zero flux




oxaloacetate
R22
coenzyme A
citrate
coenzyme A
R53r
R20
R54r
R23r
nicotinamide adenine dinucleotide
lactate
formate
ethanol
isocitrate
acetyl-coenzyme A
R55
adenosine triphosphate
acetate
acetyl-coenzyme A
R21
pyruvate
R94
R96
R24
alpha-ketoglutarate
R91
extracellular ethanol
extracellular acetate
coenzyme A
nicotinamide adenine dinucleotide
alpha-ketoglutarate
carbon dioxyde
R29r
nicotinamide adenine dinucleotide
malate
R28r
fumarate
adenosine diphosphate
adenosine diphosphate
R80
nicotinamide adenine dinucleotide
R26r
adenosine triphosphate
adenosine triphosphate
succinate
R27r
nicotinamide adenine dinucleotide
flavin adenine dinucleotide
R83
adenosine triphosphate
R81
R95
R41
pyruvate
nicotinamide adenine dinucleotide
carbon dioxyde
pyruvate
glucose-6-phosphate
R40
RR9
R9
R93
R1
adenosine triphosphate
pyruvate
adenosine diphosphate
R97r
R42
R82
carbon dioxyde
adenosine triphosphate
nicotinamide adenine dinucleotide
R10
carbon dioxyde
ribose-5-phosphate
R12r
ribulose-5-phosphate
R11r
ribose-5-phosphate
xylulose-5-phosphate
erythrose-4-phosphate
phosphoenolpyruvate
phosphoglycerate
adenosine diphosphate
adenosine triphosphate
R15r
nicotinamide adenine dinucleotide
fructose-6-phosphate
R2r
R6r
adenosine triphosphate
R3
dihydroxyacetone phosphate
adenosine diphosphate
fructose bi-phosphate
R5r
R4
biomass
adenosine diphosphate
oxaloacetate
R70
alpha-ketoglutarate
nicotinamide adenine dinucleotide
carbon dioxyde
erythrose-4-phosphate

H. pylori
genome-scale model (iIT341)



Improve Layout:

- take **group information** from SBML (if provided)
  and redraw individual pathways (make TCA circular, etc.)

- large networks: automatically **assign regions**
  for different parts of the graph (transport regions along edges etc.)

- …

Incorporate into MetaToolkit